MODULE 22

Digital Forensics Part 1 Evidence Speaks, If Handled Right

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What You Will Learn

By the end of this module, you will handle digital evidence the way courts demand - **without contaminating it**.

- The **five steps** of a forensic investigation - identification to report, in the correct order
- **Chain of custody** - the receipt of evidence that lawyers cannot tear apart
- Making a forensic image with **FTK Imager** - and why the **write blocker** is non-negotiable
- Reading **Windows Event Logs** and building a **timeline** that tells the story
- The **BSA s.63 certificate** - what makes an electronic record evidence in court

The Nirbhaya Case: Technology as the Weapon

16 December 2012, Delhi. A heinous crime in a moving bus. **Police had almost nothing** - no names, no addresses, no identity.

- CCTV of a hotel near the airport showed the bus **twice - 9:34 and 9:54 PM**. The real bus was found among hundreds
- The bus had been **washed** - yet DNA was recovered from seats, curtains and floor
- DNA matching, teeth marks and **mobile data** made the case unbreakable
- 2017: the Supreme Court said **scientific evidence had decided the fate of the guilty**

Five Steps of Forensic Investigation



- **Every professional investigation** runs these five steps - the order never changes
- Examination is the **longest step** - building the story of before, during and after the crime
- Behind every claim in your report there must be **data as evidence**

Step 1 - Identification

Finding where the evidence is. **Hurrying here is dangerous** - a missed device can mean evidence lost forever.

- Mobile, laptop, **CCTV DVR**, server, pen drive - list every device that may hold data
- Note the device's condition: **on or off**. In an ON device, RAM data is evidence that dies at switch-off
- First rule: **photograph the device in place** before touching anything
- Ask: what ran here, who touched it, what is connected - **before** you move it

Step 2 - Preservation

Keeping the evidence **exactly as it is**. From now on, not one bit will change.

- Switch off the device, **separate it from the network**, let no one touch it
- Mobile in **airplane mode or a Faraday bag** - so it cannot be wiped or changed from a distance
- Preservation means: **zero change** from this moment forward
- Every hand that touches it after this point is **recorded** - chain of custody begins here

Step 3 - Acquisition

Making an **exact copy** of the evidence. The real device is never used for investigation.

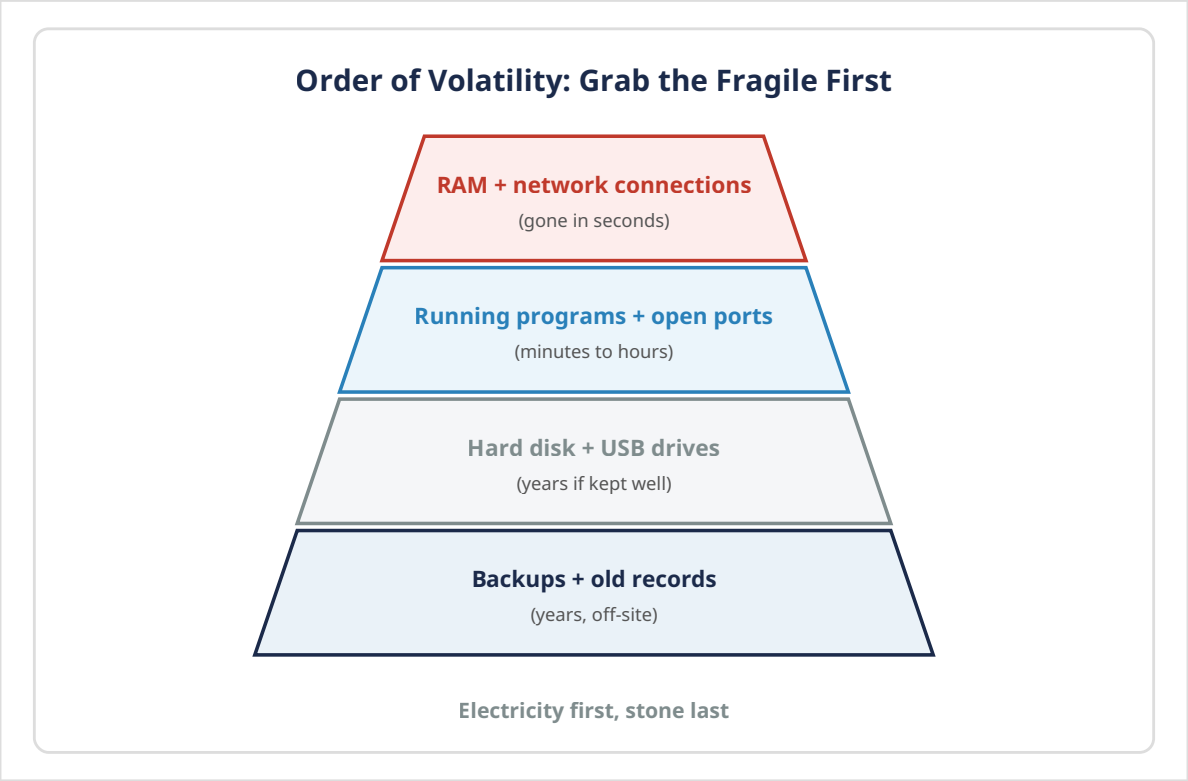
- **Imaging** = bit-by-bit copy - every bit, even 'empty' space, copied exactly
- Always connect through a **write blocker** - so nothing is written to the real data during copying
- All investigation happens on the **image**; the real drive is sealed and kept aside
- > **Real-case rule:** imaging only with legal authority + a write blocker - always.

Steps 4 and 5 - Examination and Reporting

The longest step and the final step - **finding the story, then writing it so a judge understands**.

- Examination: **deleted files, logs, timeline, photos and chats** - build what happened before, during and after
- Behind **every claim** there must be data as evidence - no guessing
- Reporting: record **what was found, how, by which methods** - presented in court
- Explain technical language in **simple words** - the judge and lawyers must understand it

Order of Volatility: Grab the Fragile First



- RAM dies with the power - **it is always first**
- Rule: pick up first **what gets wiped first**
- Backups and old records **wait for you** - RAM does not

Device ON or OFF? The First Decision

This decision is made **in seconds at the scene**. Get it wrong and evidence dies.

- Device ON: **photograph the screen**, note running programs - lift RAM data first if you have the facility and permission
- Then **separate it from the network** - before anything else
- Device OFF: **do not switch it on**. Booting changes hundreds of files instantly - contamination
- Remember: evidence on electricity comes first, **stone comes last**

Chain of Custody: The Receipt of Evidence

Chain of custody = the complete account of the evidence. Like a receipt for every rupee of a transaction.

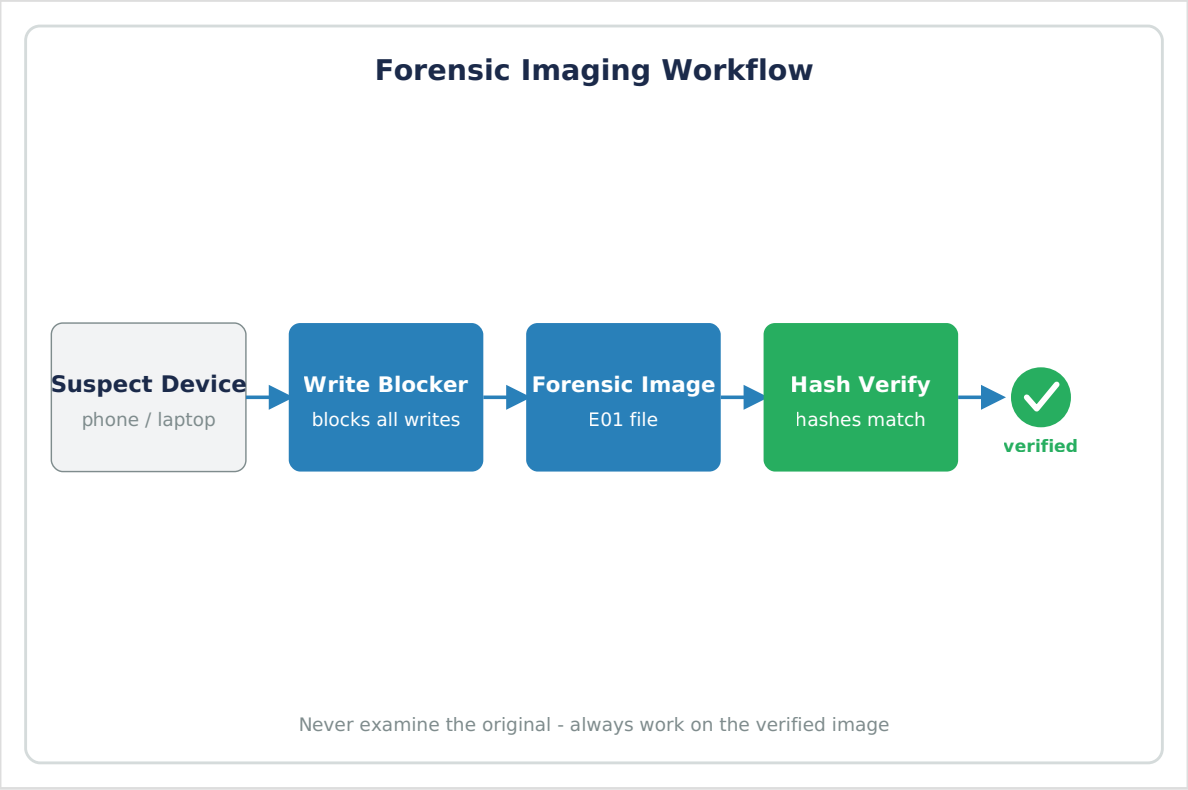
- When found, **who lifted it**, where kept, who touched it, when sent to the lab, when returned
- Rule 1: keep it **sealed** - case number, date and lifter's name written on the seal
- Rule 2: **every hand-to-hand transfer** recorded in writing, with date, time and signature
- Rule 3: store it where **no one can reach without permission** - one unrecorded hour, and the lawyer wins

Write Blocker: The One-Way Mirror

When you connect a drive, the computer **by itself writes into it** - file open times and more. In forensics, that is contamination.

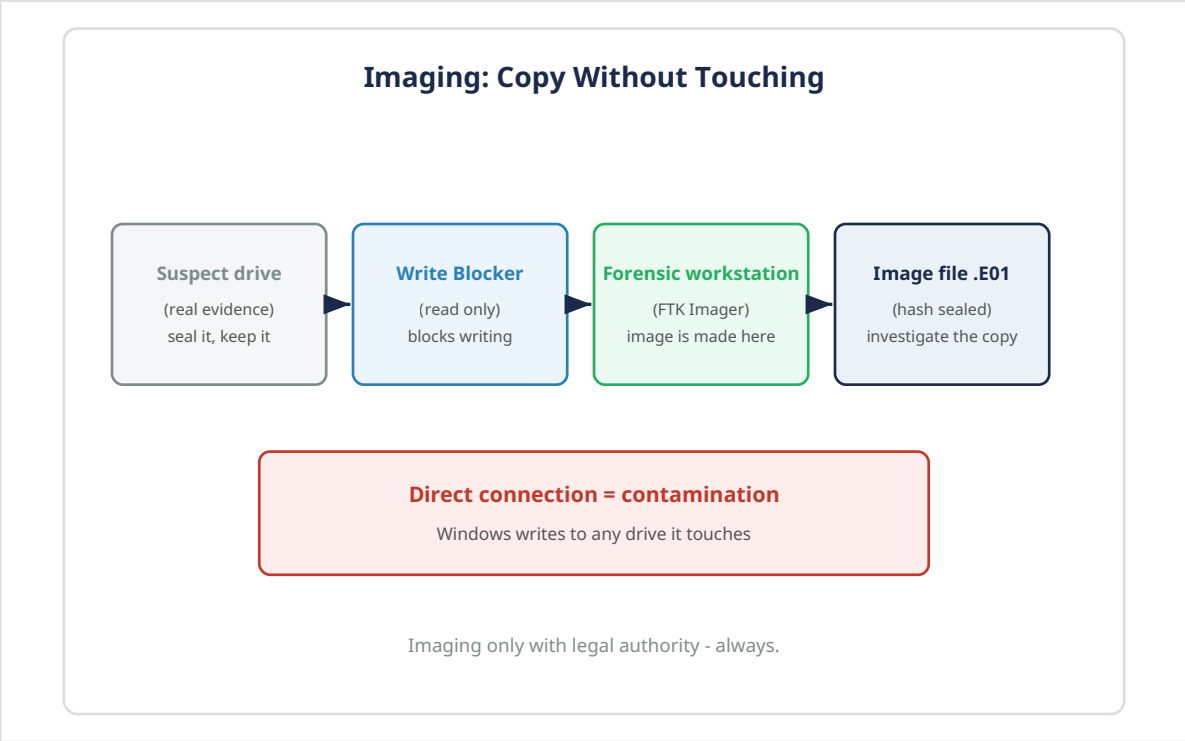
- A **write blocker** lets the drive be read, never written to - like a one-way mirror
- The suspect's drive connects **only through a write blocker** - every professional investigation
- After imaging, the **hash of drive and image must match** - proof the copy is exact
- In court, this hash **guarantees the purity** of the evidence

Imaging Workflow



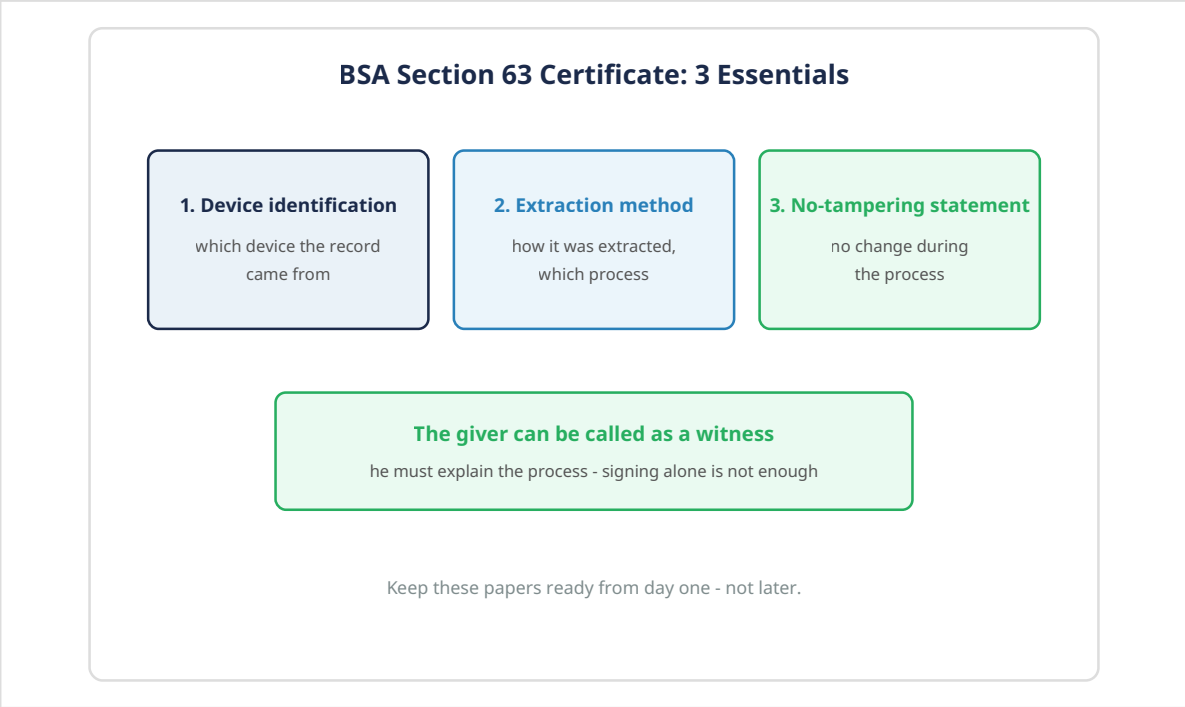
- Image = **bit-by-bit copy**, including space that looks empty
- Investigate the **image** - the real drive stays sealed
- FTK Imager records **MD5 and SHA-1 hashes** automatically

Imaging: Copy Without Touching



- Suspect drive > write blocker > workstation > **.E01 image** - the correct chain
- Without a blocker, Windows writes **access times, Recycle Bin, System Volume Information** - hash changes
- Changed hash = **tampered evidence** in the lawyer's hands

BSA Section 63: The Court Certificate



- BSA s.63 (earlier Evidence Act s.65B): an electronic record needs a **certificate** to become evidence
- The certificate needs **3 essentials**: device identification, extraction method, no-tampering statement
- The giver can be **called as a witness** - signing alone is not enough

Hash: The Seal of Evidence

The hash is the file's **digital fingerprint**. In forensics it is repeated because it is that important.

- Extract **SHA-256** at seizure, record it, extract again months later in court - match = **not one bit changed**
- FTK Imager extracts **MD5 and SHA-1** together - both matching almost removes the possibility of tampering
- **Hash collision:** two different files, same hash - possible in MD5, so **MD5 alone is not relied upon**
- Both hashes in court make challenging the evidence's purity **very difficult**

Windows Event IDs: The OS Diary

Windows writes every important event in its logs. **Event Viewer** is the investigator's treasure chest.

Event ID	Meaning	What it tells you
4624	Successful logon	Who logged in, when, and how (logon type)
4625	Failed logon	Wrong-password attempts - guessing attacks show here
4634	Logoff	When the user exited the session
4672 / 4648	Special rights / run-as	Admin-level logins - a midnight 4672 is worth investigating

Logon Types: How Did They Get In?

Event 4624's **Logon Type** tells HOW the login happened. Four types matter to you:

- **Type 2 - Interactive:** logged in sitting in front of the computer
- **Type 3 - Network:** connected over the network - like opening a shared folder
- **Type 10 - RemoteInteractive:** Remote Desktop from far away - a **midnight Type 10 is a red flag**
- **Type 11 - CachedInteractive:** logged in from an old record without network
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> **Lab 3:** open Event Viewer on your own computer and read these events yourself.

Timeline Analysis: One Story from Many Marks

The most powerful weapon of forensic investigation - **placing all digital marks in the order of time**.

- Event logs, browser history, call records, CCTV timestamps - **all in one chart**
- The Nirbhaya timeline: **9:34 PM bus seen, 9:54 PM bus seen again** - ordered marks told the story
- A claim of 'I was home at 10' **dies** when the phone connected to the crime-scene tower at 10
- Start with a **spreadsheet**: date-time, event, evidence source - judges understand timelines

Real Case: The Nirbhaya Investigation

How technology caught criminals the police **could not even name** at the start:

- Hotel CCTV near the airport: the bus seen at **9:34 and 9:54 PM** - identified among hundreds of buses
- The bus had been **washed** - yet DNA found on seats, curtains and floor
- **Mobile tower records** placed the accused on the same route at the same time
- Caught within days; the chain of scientific evidence **held till the Supreme Court**
- Source The Week, FirstPost

Real Case: Lessons for the Investigator

Three lessons from the Nirbhaya case - **for your next investigation**:

- Collecting evidence is **half the battle** - handling it right (hash, chain of custody) is the other half
- The **timeline turns scattered marks into a story** - CCTV time, tower records, DNA, all in one order
- In court only evidence holds **whose every step is accounted for** - correct legal process from day one
- The criminal tried to erase the marks - **science caught him anyway**. Never assume a cleaned place holds nothing
- Source The Week, The Bridge Chronicle

Common Mistakes That Kill Evidence

Most digital evidence is destroyed by **well-meaning officers in the first hour** - not by criminals.

- **Switching ON** a seized device 'just to check' - changes hundreds of timestamps instantly
- **Booting an OFF device** at the scene - the system contaminates itself on startup
- Connecting a suspect drive **without a write blocker** - Windows writes, hash changes, lawyer wins
- Leaving **one unrecorded hour** in the chain of custody - the gap becomes the defence's weapon

First-Response Steps: The Golden Order

At the scene, in this order - **photograph, isolate, seize, seal, document**:

- **Photograph** every device in place - screen, connections, condition
- **Isolate:** flight mode or Faraday bag for mobiles; disconnect computers from the network
- **Seize and seal** - case number, date, your name on the seal
- Start the **chain of custody log** immediately - hash at the earliest possible moment

Your Turn: Labs in This Module

Theory without practice is trivia. Chapter 22 has **four hands-on labs** - all on your own devices:

- **Lab 1:** Make a forensic image of your own USB drive with FTK Imager (.E01)
- **Lab 2:** Verify the image hash and fill a chain of custody log
- **Lab 3:** Read logon events (4624, 4625, 4634) in Windows Event Viewer
- **Lab 4:** Draw the write-blocker acquisition chain - and list what Windows writes without one
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> **Safety:** your own devices only. Real-case imaging needs legal authority + a write blocker.

Key Takeaways

If you remember nothing else from Module 22, remember this:

- Five steps, fixed order: **identify, preserve, acquire, examine, report**
- Volatile first: **RAM dies with the power** - electricity first, stone last
- **Never investigate the real device** - image it through a write blocker
- Chain of custody is **the receipt of evidence** - no gaps, ever
- The **hash is the seal** - SHA-256 recorded at seizure, matched in court
- BSA s.63 certificate + timeline = **evidence a court can trust**

Review Questions

- 1. You find a laptop ON at a crime scene, showing an open chat. List your first five actions in order - and say why each matters.
- 2. The chain of custody log has a 6-hour gap between the station malkhana and the lab. The defence lawyer attacks the evidence. What exactly is his argument?
- 3. An officer images a suspect's drive by connecting it directly to his computer. Name three things Windows writes to it - and explain what happens to the hash.
- 4. Event Viewer shows a 4624 with Logon Type 10 at 2 AM, followed by several 4625 events. What story does this tell?
- 5. Under BSA s.63, what three things must the certificate contain - and why can the giver be called as a witness?

NEXT: MODULE 23

Digital Forensics Part 2 Photos, Video and Browsers Testify

Metadata, image forensics, browser artefacts - the silent witnesses

Cyber Crime Investigator - Training Manual